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FPT INTERNATIONAL TRAINING INSTITUTE FPT – APTECH COMPUTER EDUCATION

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PROJECT REPORT HOPSONTAI CLINIC MANAGEMENT SYSTEM

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Building the Hopsontai Clinic Management System has been a true team effort, and we sincerely appreciate everyone who contributed to bringing this project to fruition.

SYNOPSIS

The Hopsontai Clinic Management System is a modern, full-stack web application designed to digitize and streamline the day-to-day operations of a general and traditional-medicine clinic. Built on the MERN-style stack (MongoDB, Express, React, and Node.js), the system replaces fragmented, paper-based workflows with a single, role-driven platform accessible from any web browser.

The application serves five distinct user roles – Administrator, Doctor, Customer-Care Staff, Accountant, and Patient – each with a dedicated portal and permissions enforced through JSON Web Token (JWT) authentication and role-based access control (RBAC). Patients can browse specialist departments, read health articles, and book appointments online; doctors manage their schedules, record medical examinations, and issue prescriptions; staff confirm appointments and handle customer inquiries; accountants generate and settle invoices; and administrators oversee the entire system, including users, departments, doctor schedules, medicines, and published content.

Beyond core management features, the platform offers patient-facing value such as a public website with department and specialist listings, a clinical blog (CMS), a quick-booking widget, a contact form, and an AI assistant for the administrator. By centralizing patient records, appointments, prescriptions, and billing into one consistent database, the system improves data accuracy, enhances service quality, and supports better, faster decision-making for clinic management.

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Chapter 1. PROBLEM DEFINITION

The development of the Hopsontai Clinic Management System is driven by the need to modernize clinic operations and overcome the limitations of manual, paper-based and disconnected management practices. Based on the project context, the core problems addressed are:

- **Inefficient manual appointment handling:** Booking by phone or walk-in leads to double-booking, long waiting times, and lost records, making it hard to balance doctor workload across departments and time slots.
- **Fragmented and unsafe patient data:** Without a centralized database, patient profiles, medical histories, prescriptions, and invoices are scattered across paper files or spreadsheets, causing data inconsistency, duplication, and risk of loss.
- **Difficult clinical record keeping:** Doctors struggle to retrieve a patient's previous diagnoses, vital signs, and prescribed medicines quickly during an examination, which affects the quality and continuity of care.
- **Manual and error-prone billing:** Calculating consultation fees and pharmacy charges by hand is slow and prone to mistakes, while tracking which invoices are paid or unpaid becomes increasingly difficult as patient volume grows.
- **Lack of real-time oversight and analytics:** Clinic managers have no consolidated view of revenue, appointment volume, or department performance, making strategic decisions difficult.
- **Limited online presence and patient engagement:** Without a public website, patients cannot easily discover specialists, read trustworthy health information, or book appointments online, reducing the clinic's reach and convenience.
- **Inconsistent access control:** A single shared system without role separation risks exposing sensitive medical and financial data to unauthorized staff.

The Hopsontai Clinic Management System solves these problems by providing a centralized, secure, role-based web platform that manages the entire clinic workflow – from online appointment booking to examination, prescription, billing, and reporting – within a single, consistent application.

Chapter 2. CUSTOMER'S REQUIREMENTS SPECIFICATIONS (CRS)

2.1 Business / Project Objective

The main objective of the Hopsontai Clinic Management System project is to develop a comprehensive and efficient web application that supports the management and operation of a multi-specialty clinic. The application aims to streamline daily clinical and administrative activities, improve data accuracy, and enhance decision-making through effective information management and reporting.

The specific objectives of the project are as follows:

User & Access Management

- Manage user accounts for five roles: Administrator, Doctor, Staff, Accountant, and Patient.
- Authenticate users securely with hashed passwords (bcrypt) and JSON Web Tokens (JWT).
- Enforce role-based access control so each role only accesses its permitted features.
- Allow patients to self-register and existing staff/doctors to be provisioned by the administrator.

Patient & Profile Management

- Store and manage patient information including full name, date of birth, gender, identity card (CCCD/CMND), phone number, address, insurance code (BHYT), and emergency contact.
- Maintain doctor profiles with specialization, department, qualifications, years of experience, consultation fee, and biography.
- Maintain staff profiles for customer-care and accounting personnel.

Appointment & Scheduling Management

- Allow administrators to define departments and create doctor work schedules with date, time blocks, and maximum patient capacity.
- Allow patients to book appointments online by selecting a department, doctor, date, and time slot.
- Allow staff/doctors to confirm, complete, or cancel appointments, automatically tracking booked capacity per schedule.
- Provide a public quick-booking widget for fast appointment requests by name and phone.

Clinical Examination & Prescription Management

- Allow doctors to record medical records with vital signs (height, weight, blood pressure, heart rate, temperature), diagnosis, and clinical notes.
- Allow doctors to issue prescriptions linking medicines with dosage, frequency, duration, and special instructions.
- Maintain a medicine catalog with code, name, active ingredient, usage route, unit, price, and stock quantity, automatically deducting stock when dispensed.

Billing & Payment Management

- Generate consultation and pharmacy invoices linked to appointments and patients.

- Track invoice status (Unpaid, Paid, Refunded) and record the staff member who processed each payment.
- Store invoice details for dispensed medicines, including quantity, unit price, and subtotal.

Content Management & Communication

- Publish clinical blog posts (CMS) with title, slug, thumbnail, rich-text content, and publication status.
- Receive and manage contact inquiries submitted from the public website.
- Provide an AI assistant in the admin dashboard to answer clinic-related queries.

Reporting & Analytics

- Provide an administrator dashboard with statistics on users, appointments, revenue, and department activity.
- Expose public statistics (e.g., number of doctors, departments) for the website.
- Support data-driven decision-making through consolidated reporting.

Input to the System

Users need to provide:

- Account information: username (phone number), password, email, role.
- Patient information: full name, date of birth, gender, identity card, phone number, address, insurance code, emergency contact.
- Doctor information: full name, specialization, department, qualifications, experience, consultation fee, biography.
- Appointment information: patient, department, doctor, requested date/time, symptoms.
- Clinical information: vital signs, diagnosis, clinical notes, prescribed medicines.
- Medicine information: code, name, active ingredient, usage route, unit, price, stock quantity.
- Content information: post title, content, thumbnail, status; contact inquiries.

Output from the System

- Patient portal: profile, upcoming and past appointments, medical history, prescriptions, and invoices.
- Doctor portal: daily schedule, patient list, examination forms, and prescription records.
- Staff portal: appointment confirmation queue and contact-inquiry management.
- Accountant portal: invoice list, payment processing, and revenue figures.
- Administrator portal: full management of users, departments, schedules, medicines, posts, and analytics dashboards.
- Public website: home page, department and specialist listings, service packages, blog posts, and contact page.

List of Deliverables

- Project Report (Word Document)
- User Guide

- Source Code (Frontend + Backend)
- Database seed script with sample data

2.2 Hardware / Software Requirements

Hardware

- A computer capable of running a modern web browser and Node.js (Intel Core i3 / equivalent or better recommended).
- Minimum 4 GB of RAM (8 GB recommended for running the database, backend, and frontend simultaneously).
- At least 2 GB of free disk space for the source code, dependencies, and database.

Software

- Operating System: Windows 10/11, macOS, or Linux.
- Runtime: Node.js (v18 or higher) and npm.
- Database: MongoDB (local installation or MongoDB Atlas cloud).
- Frontend: React 18 with Vite, React Router, Axios, SweetAlert2.
- Backend: Express 4, Mongoose 7, jsonwebtoken, bcryptjs, cors, dotenv.
- Tools: Visual Studio Code, Git, MongoDB Compass, a modern web browser (Chrome / Edge / Firefox).

2.3 Architecture and Design of the Project

The application follows a client-server, three-tier web architecture, separating presentation, application logic, and data storage:

- **Presentation tier (Client):** A Single-Page Application (SPA) built with React and Vite. It renders role-specific layouts (Public, Patient, Doctor, Admin) and communicates with the backend exclusively through RESTful JSON APIs using Axios.
- **Application tier (Server):** A RESTful API built with Node.js and Express, organized into domain-driven modules (auth, profiles, scheduling, booking, clinical, billing, cms). It handles business logic, authentication (JWT), authorization (RBAC middleware), and validation.
- **Data tier (Database):** A MongoDB database accessed through Mongoose ODM. Sixteen collections store all clinic data, with references (ObjectId) modeling relationships between entities.

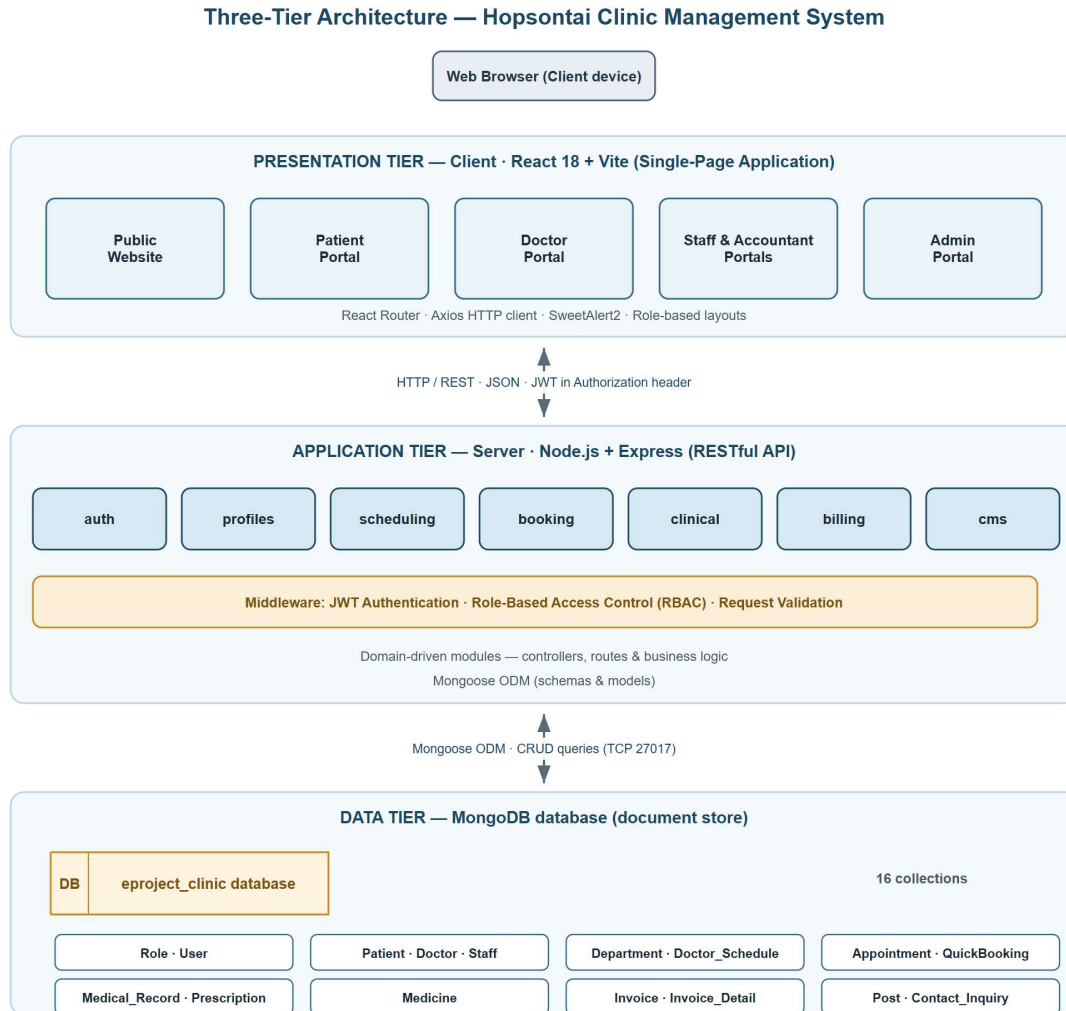


Figure 1: System architecture (three-tier) overview

Communication between tiers is stateless: each authenticated request carries a JWT in the Authorization header, which the backend verifies before granting access to protected resources. This design promotes scalability, maintainability, and a clear separation of concerns.

Chapter 3. SCOPE OF THE WORK (IN BRIEF)

The scope of work for the Hopsontai Clinic Management System project is defined as follows.

3.1 User Interface & Public Website

The system provides a responsive, user-friendly interface built with React. The public website includes a home page, department listing, specialist (doctor) listing, service packages, a clinical blog, an about page, and a contact page, allowing visitors to learn about the clinic and book appointments without logging in.

3.2 Authentication & Role-Based Access

The system supports secure login and registration. Five roles – Admin, Doctor, Staff, Accountant, and Patient – are authenticated via JWT, with middleware enforcing role-based access to every protected route. An admin impersonation feature supports troubleshooting.

3.3 Patient & Profile Management

The system manages patient profiles (identity card, insurance, contact, emergency contact) and professional profiles for doctors and staff. Patients can update their own profiles; administrators manage all accounts.

3.4 Appointment & Scheduling

Administrators define departments and doctor schedules (date, time block, capacity). Patients book appointments online; staff and doctors confirm, complete, or cancel them. The system tracks booked capacity to prevent overbooking, and offers a public quick-booking widget.

3.5 Clinical Examination & Prescriptions

Doctors record medical examinations (vital signs, diagnosis, clinical notes) tied to each appointment and issue prescriptions from the medicine catalog, with automatic stock deduction. Patients can review their medical history and prescriptions.

3.6 Billing & Payment

Accountants generate consultation and pharmacy invoices, process payments, and track invoice status. Invoice details record each dispensed medicine with quantity, unit price, and subtotal.

3.7 Content Management & Communication

Administrators publish and manage clinical blog posts (CMS) and review contact inquiries submitted from the website. Staff can also handle inquiries. An AI assistant supports the admin with clinic queries.

3.8 Reporting, Integration & Extensibility

The administrator dashboard consolidates statistics on users, appointments, and revenue. The modular backend architecture (domain modules + REST APIs) allows future enhancements such as payment-gateway integration, SMS/email notifications, or a mobile client.

TASK SHEET REVIEW 1

Project Title: Hopsontai Clinic Management System

Date of Preparation of Activity Plan: 06-Jun-2026

Sr. No.	Task	Actual Days	Teammate	Status
1	Acknowledgement & Synopsis	1	Phan Minh Tai	Completed
2	Problem Definition	1	Tran Dang Khoa	Completed
3	Customer Requirement Specification (CRS)	1	Tran Dang Khoa	Completed
4	Scope of Work	1	Nguyen Ngoc Son	Completed
5	Hardware / Software Requirements	1	Nguyen Ngoc Son	Completed
6	Architecture & Design of the Project	1	Nguyen Van Hop	Completed
7	Task Sheet	1	Phan Minh Tai	Completed

Signature of Instructor:

Mrs. Pham Thi Lanh

Signature of Team Leader:

Phan Minh Tai

Chapter 4. DATABASE DESIGN

4.1 Relational Model of the Database

The Hopsontai Clinic Management System uses a MongoDB (document) database accessed through Mongoose. Although MongoDB is non-relational, the schema is designed in a normalized, relational style: each entity is stored in its own collection and relationships are modeled using ObjectId references. The database consists of sixteen collections, summarized in the relationship overview below.

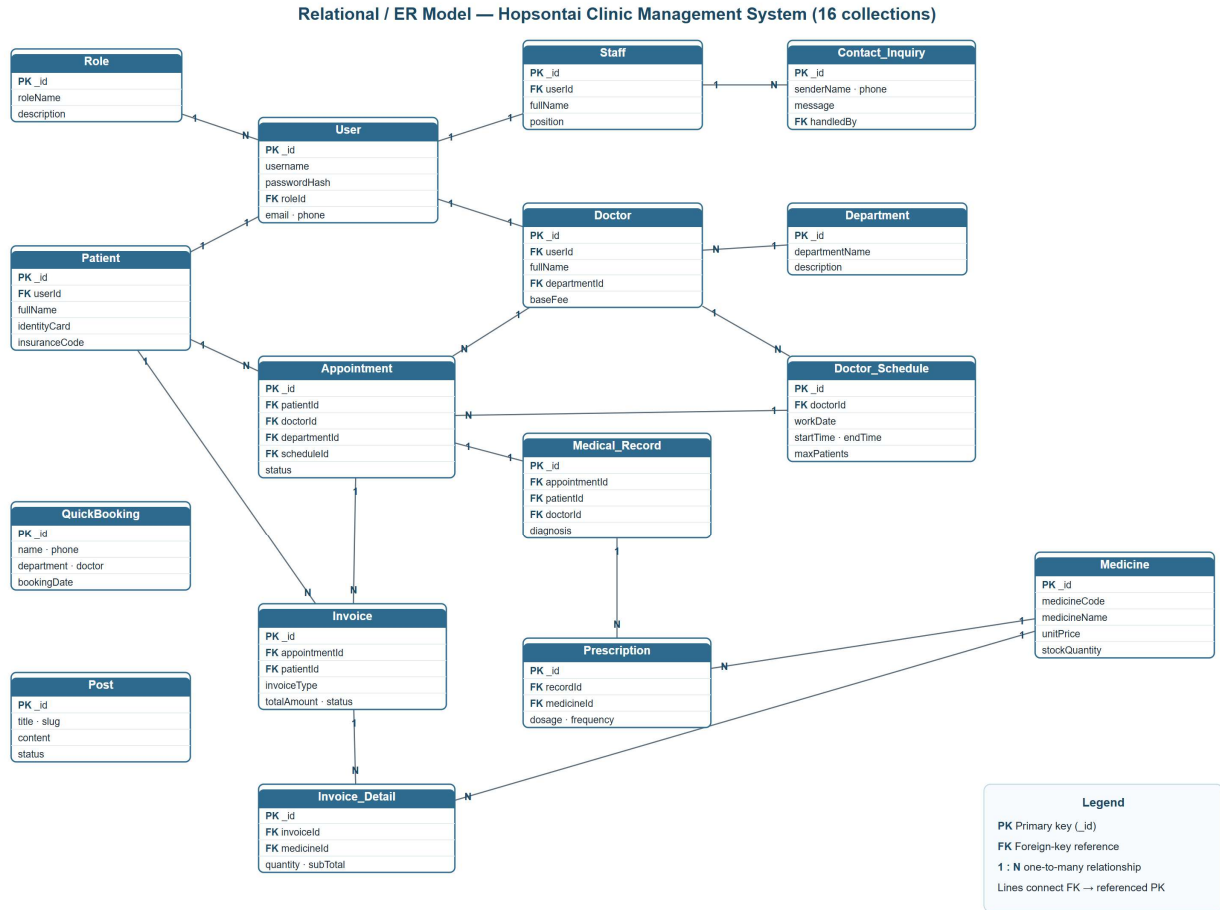


Figure 2: Relational / ER model of the database

Key relationships between collections

Relationship	Type	Description
User → Role	N : 1	Each user is assigned exactly one role.
Patient/Doctor/Staff → User	1 : 1	Each profile is linked to one user account.
Doctor → Department	N : 1	Each doctor belongs to one department.
Doctor_Schedule → Doctor	N : 1	A doctor has many work schedules.
Appointment → Patient / Doctor / Department / Schedule	N : 1	An appointment references the patient, doctor, department, and schedule.
Medical_Record → Appointment / Patient / Doctor	1 : 1 / N : 1	One medical record per appointment.

Relationship	Type	Description
Prescription → Medical_Record / Medicine	N : 1	A record may have many prescription lines.
Invoice → Appointment / Patient	N : 1	Consultation and pharmacy invoices per appointment.
Invoice_Detail → Invoice / Medicine	N : 1	Line items for pharmacy invoices.
Contact_Inquiry → Staff	N : 1	Inquiry handled by a staff member.

4.2 Data Description of Tables

Each collection is described below. The implicit Mongoose timestamp fields (createdAt, updatedAt) and the auto-generated primary key (_id) are included where relevant. “Key” denotes PK (primary key) or FK (foreign-key reference).

Table 1: Role collection data structure

Field name	Data Type	Null	Key	Describe	Notes
_id	ObjectId	Not null	PK	Role identifier	Auto-generated
roleName	String	Not null		Role name	enum: admin, doctor, staff, accountant, patient · unique
description	String	Null		Role description	
createdAt / updatedAt	Date	Not null		Record timestamps	Auto (timestamps)

Table 2: User collection data structure

Field name	Data Type	Null	Key	Describe	Notes
_id	ObjectId	Not null	PK	User identifier	Auto-generated
username	String	Not null		Login username	Phone number · unique
passwordHash	String	Not null		Hashed password	bcrypt (auto-hashed)
roleId	ObjectId	Not null	FK	Role reference	→ Role
email	String	Null		Email address	unique (sparse)
phone	String	Null		Phone number	unique (sparse)
isActive	Boolean	Null		Account active	default: true
isRegistered	Boolean	Null		Self-registered flag	default: false
lastLoginAt	Date	Null		Last login time	
createdAt / updatedAt	Date	Not null		Record timestamps	Auto

Table 3: Patient collection data structure

Field name	Data Type	Null	Key	Describe	Notes
_id	ObjectId	Not null	PK	Patient identifier	Auto-generated

Field name	Data Type	Null	Key	Describe	Notes
userId	ObjectId	Not null	FK	User reference	→ User · unique
fullName	String	Not null		Patient full name	
avatarURL	String	Null		Profile photo URL	
dateOfBirth	Date	Not null		Date of birth	
gender	String	Not null		Gender	enum: Nam, Nữ, Khác
identityCard	String	Not null		Citizen ID (CCCD/CMND)	unique
phoneNumber	String	Not null		Phone number	unique
address	String	Null		Home address	
insuranceCode	String	Null		Health insurance code (BHYT)	unique (sparse)
emergencyContact	String	Null		Emergency contact	
createdAt / updatedAt	Date	Not null		Record timestamps	Auto

Table 4: Doctor collection data structure

Field name	Data Type	Null	Key	Describe	Notes
_id	ObjectId	Not null	PK	Doctor identifier	Auto-generated
userId	ObjectId	Not null	FK	User reference	→ User · unique
fullName	String	Not null		Doctor full name	
avatarURL	String	Null		Profile photo URL	
specialization	String	Not null		Specialization	
departmentId	ObjectId	Not null	FK	Department reference	→ Department
qualifications	String	Null		Degrees / qualifications	
experienceYears	Number	Null		Years of experience	default: 0
baseFee	Number	Null		Consultation fee	
bio	String	Null		Biography	
isActive	Boolean	Null		Active status	default: true
createdAt / updatedAt	Date	Not null		Record timestamps	Auto

Table 5: Staff collection data structure

Field name	Data Type	Null	Key	Describe	Notes
_id	ObjectId	Not null	PK	Staff identifier	Auto-generated

Field name	Data Type	Null	Key	Describe	Notes
userId	ObjectId	Not null	FK	User reference	→ User · unique
fullName	String	Not null		Staff full name	
avatarURL	String	Null		Profile photo URL	
phoneNumber	String	Null		Phone number	
position	String	Null		Job position	e.g. Customer Care, Accountant
createdAt / updatedAt	Date	Not null		Record timestamps	Auto

Table 6: Department collection data structure

Field name	Data Type	Null	Key	Describe	Notes
_id	ObjectId	Not null	PK	Department identifier	Auto-generated
departmentName	String	Not null		Department / specialty name	unique
description	String	Null		Department description	
createdAt / updatedAt	Date	Not null		Record timestamps	Auto

Table 7: Doctor_Schedule collection data structure

Field name	Data Type	Null	Key	Describe	Notes
_id	ObjectId	Not null	PK	Schedule identifier	Auto-generated
doctorId	ObjectId	Not null	FK	Doctor reference	→ Doctor
workDate	Date	Not null		Working date	
startTime	String	Not null		Start time	Format HH:mm (e.g. 08:00)
endTime	String	Not null		End time	Format HH:mm (e.g. 12:00)
maxPatients	Number	Not null		Maximum patients	
currentBooked	Number	Null		Currently booked count	default: 0
status	String	Null		Schedule status	enum: Available, Full, Canceled
createdAt / updatedAt	Date	Not null		Record timestamps	Auto

Table 8: Appointment collection data structure

Field name	Data Type	Null	Key	Describe	Notes
_id	ObjectId	Not null	PK	Appointment identifier	Auto-generated
patientId	ObjectId	Not null	FK	Patient reference	→ Patient

Field name	Data Type	Null	Key	Describe	Notes
requestedDate	Date	Not null		Requested date	
requestedTime	String	Null		Requested time	
symptoms	String	Null		Reason / symptoms	
departmentId	ObjectId	Null	FK	Department reference	→ Department
doctorId	ObjectId	Null	FK	Doctor reference	→ Doctor
scheduleId	ObjectId	Null	FK	Schedule reference	→ Doctor_Schedule
status	String	Null		Appointment status	enum: Pending, Confirmed, Completed, Canceled
confirmedBy	ObjectId	Null	FK	Confirming user	→ User
createdAt / updatedAt	Date	Not null		Record timestamps	Auto

Table 9: Medicine collection data structure

Field name	Data Type	Null	Key	Describe	Notes
_id	ObjectId	Not null	PK	Medicine identifier	Auto-generated
medicineCode	String	Not null		Medicine code	unique
medicineName	String	Not null		Medicine name	
activeIngredient	String	Null		Active ingredient	
usageRoute	String	Null		Route of administration	enum: Uống, Bôi, Tiêm
unit	String	Not null		Unit	e.g. tablet
unitPrice	Number	Not null		Unit price (VND)	
stockQuantity	Number	Null		Stock quantity	default: 0
isActive	Boolean	Null		Active status	default: true
createdAt / updatedAt	Date	Not null		Record timestamps	Auto

Table 10: Medical_Record collection data structure

Field name	Data Type	Null	Key	Describe	Notes
_id	ObjectId	Not null	PK	Record identifier	Auto-generated
appointmentId	ObjectId	Not null	FK	Appointment reference	→ Appointment · unique
patientId	ObjectId	Not null	FK	Patient reference	→ Patient
doctorId	ObjectId	Not null	FK	Doctor reference	→ Doctor
height	Number	Null		Height (cm)	
weight	Number	Null		Weight (kg)	

Field name	Data Type	Null	Key	Describe	Notes
bloodPressure	String	Null		Blood pressure	e.g. 120/80
heartRate	Number	Null		Heart rate (bpm)	
temperature	Number	Null		Body temperature (°C)	
diagnosis	String	Not null		Diagnosis	
clinicalNotes	String	Null		Clinical notes / advice	
createdAt / updatedAt	Date	Not null		Record timestamps	Auto

Table 11: Prescription collection data structure

Field name	Data Type	Null	Key	Describe	Notes
_id	ObjectId	Not null	PK	Prescription identifier	Auto-generated
recordId	ObjectId	Not null	FK	Medical record reference	→ Medical_Record
medicineId	ObjectId	Not null	FK	Medicine reference	→ Medicine
quantity	Number	Not null		Quantity prescribed	
dosage	String	Not null		Dosage	e.g. 500mg
frequency	String	Not null		Frequency	e.g. 2 times/day
durationDays	Number	Not null		Duration (days)	
specialInstructions	String	Null		Special instructions	
createdAt / updatedAt	Date	Not null		Record timestamps	Auto

Table 12: Invoice collection data structure

Field name	Data Type	Null	Key	Describe	Notes
_id	ObjectId	Not null	PK	Invoice identifier	Auto-generated
appointmentId	ObjectId	Not null	FK	Appointment reference	→ Appointment
patientId	ObjectId	Not null	FK	Patient reference	→ Patient
invoiceType	String	Not null		Invoice type	enum: Consultation, Pharmacy
totalAmount	Number	Not null		Total amount (VND)	
status	String	Null		Payment status	enum: Unpaid, Paid, Refunded
issuedAt	Date	Null		Issue date	default: now
paidAt	Date	Null		Payment date	

Field name	Data Type	Null	Key	Describe	Notes
processedBy	ObjectId	Null	FK	Processing user (cashier)	→ User
createdAt / updatedAt	Date	Not null		Record timestamps	Auto

Table 13: Invoice_Detail collection data structure

Field name	Data Type	Null	Key	Describe	Notes
_id	ObjectId	Not null	PK	Detail identifier	Auto-generated
invoiceId	ObjectId	Not null	FK	Invoice reference	→ Invoice
medicineId	ObjectId	Not null	FK	Medicine reference	→ Medicine
quantity	Number	Not null		Quantity	
unitPrice	Number	Not null		Unit price (VND)	
subTotal	Number	Not null		Line subtotal (VND)	
createdAt / updatedAt	Date	Not null		Record timestamps	Auto

Table 14: QuickBooking collection data structure

Field name	Data Type	Null	Key	Describe	Notes
_id	ObjectId	Not null	PK	Quick-booking identifier	Auto-generated
name	String	Not null		Customer name	
phone	String	Not null		Customer phone	
department	String	Null		Requested department	
doctor	String	Null		Requested doctor	
bookingDate	String	Not null		Booking date	Format YYYY-MM-DD
time	String	Null		Booking time	e.g. 09:00
createdAt / updatedAt	Date	Not null		Record timestamps	Auto

Table 15: Post collection data structure

Field name	Data Type	Null	Key	Describe	Notes
_id	ObjectId	Not null	PK	Post identifier	Auto-generated
title	String	Not null		Post title	
slug	String	Not null		URL slug	unique
thumbnailURL	String	Null		Cover image URL	

Field name	Data Type	Null	Key	Describe	Notes
content	String	Not null		Rich-text content (HTML)	
status	String	Null		Publication status	enum: Draft, Published, Archived
publishedAt	Date	Null		Publication date	
createdAt / updatedAt	Date	Not null		Record timestamps	Auto

Table 16: Contact_Inquiry collection data structure

Field name	Data Type	Null	Key	Describe	Notes
_id	ObjectId	Not null	PK	Inquiry identifier	Auto-generated
senderName	String	Not null		Sender name	
senderPhone	String	Not null		Sender phone	
senderEmail	String	Null		Sender email	lowercase
message	String	Not null		Inquiry message	
isResolved	Boolean	Null		Resolved flag	default: false
handledBy	ObjectId	Null	FK	Handling staff	→ Staff
submittedAt	Date	Null		Submission date	default: now
createdAt / updatedAt	Date	Not null		Record timestamps	Auto

Chapter 5. DFD DIAGRAMS

Data Flow Diagrams (DFD) illustrate how data moves through the Hopsontai Clinic Management System between external entities, processes, and data stores. The Context Diagram (Level 0) presents the system as a single process interacting with its external entities, while Level 1 diagrams decompose the system into its major functional modules.

5.1 Context DFD (Level 0)

The context diagram shows the Hopsontai Clinic Management System as a central process interacting with five external entities: Patient, Doctor, Staff, Accountant, and Administrator. Each entity sends inputs to and receives outputs from the system.

Context Diagram (DFD Level 0) — Hopsontai Clinic Management System

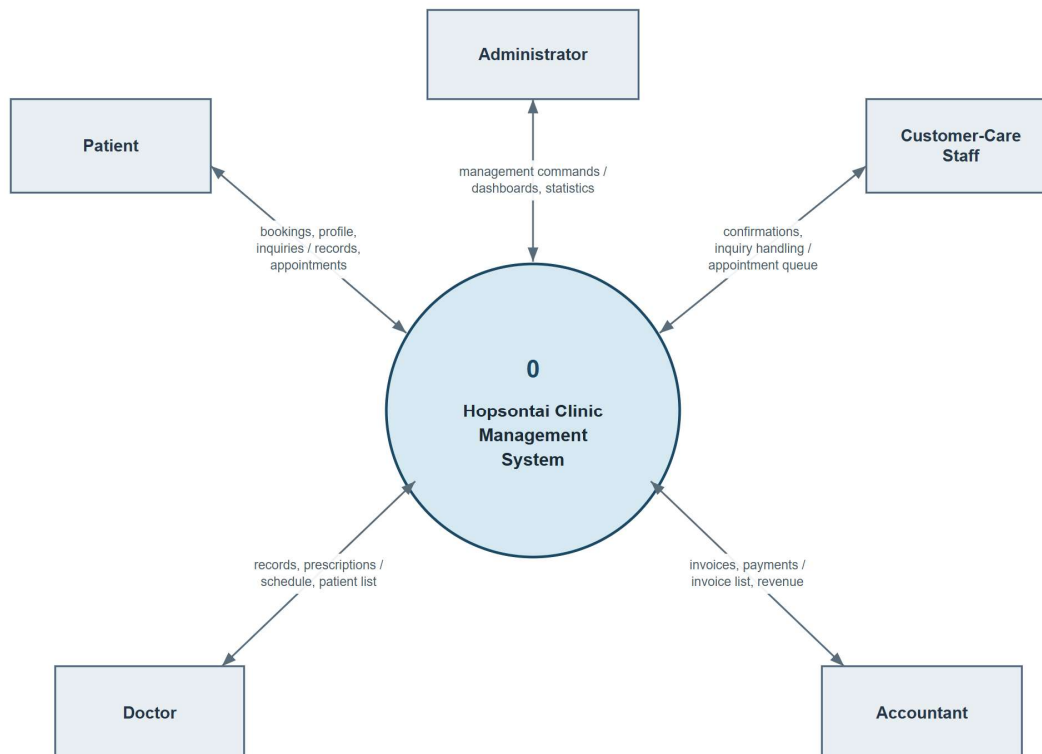


Figure 3: Context DFD (Level 0)

External entities and primary data flows

External Entity	Input to System	Output from System
Patient	Registration, login, appointment booking, profile update, contact inquiry	Profile, appointments, medical history, prescriptions, invoices
Doctor	Login, medical record, prescription, appointment update	Daily schedule, patient list, examination & prescription data

External Entity	Input to System	Output from System
Staff	Login, appointment confirmation, inquiry handling	Appointment queue, contact inquiries
Accountant	Login, invoice creation, payment processing	Invoice list, payment status, revenue figures
Administrator	Login, manage users / departments / schedules / medicines / posts	Dashboards, statistics, full management data

5.2 DFD Level 1 – Functional Modules

The system is decomposed into the following major processes, each backed by one or more data stores (MongoDB collections).

5.2.1 Authentication & Account

Handles registration, login, JWT issuance, and role-based authorization. Data stores: User, Role.

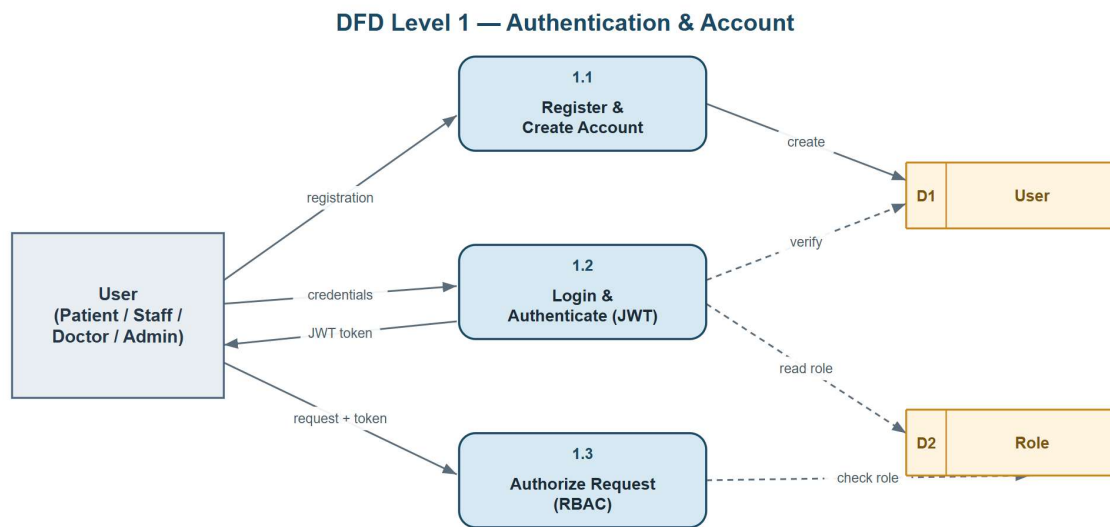


Figure 4: DFD Level 1 – Authentication & Account

5.2.2 Appointment & Scheduling

Manages departments, doctor schedules, appointment booking, confirmation, and capacity tracking. Data stores: Department, Doctor_Schedule, Appointment, QuickBooking.

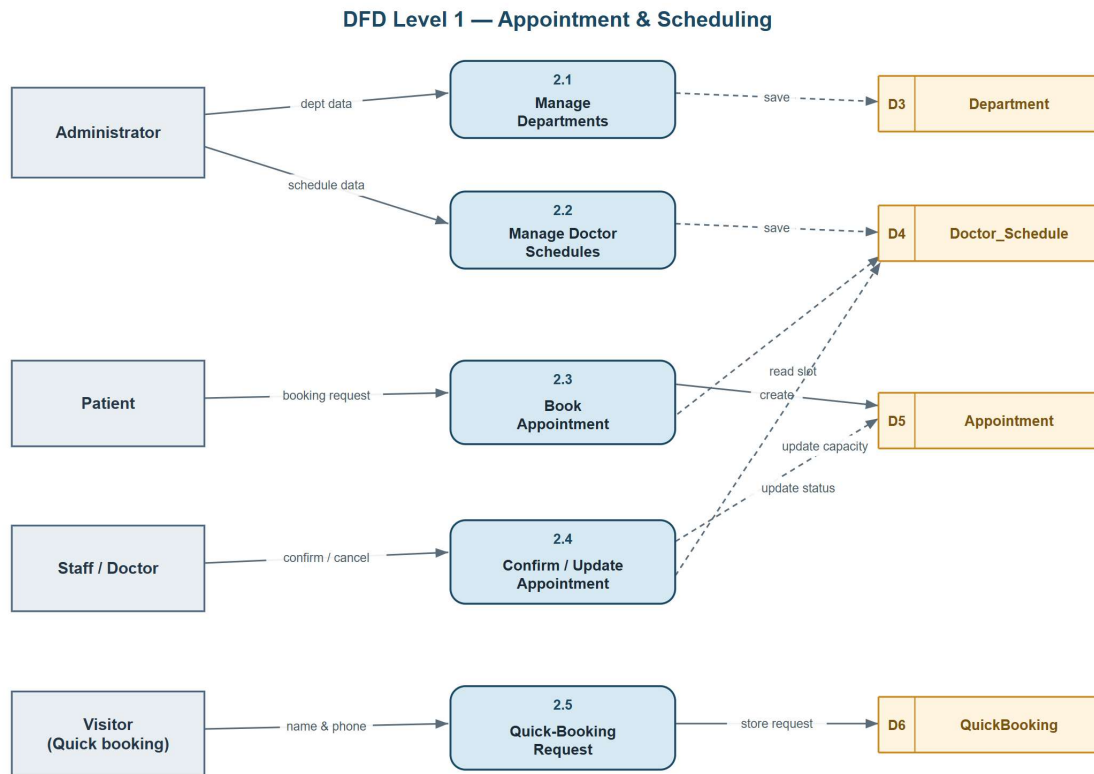


Figure 5: DFD Level 1 – Appointment & Scheduling

5.2.3 Clinical Examination & Prescription

Records medical examinations and prescriptions, and updates medicine stock. Data stores: Medical_Record, Prescription, Medicine.

DFD Level 1 — Clinical Examination & Prescription

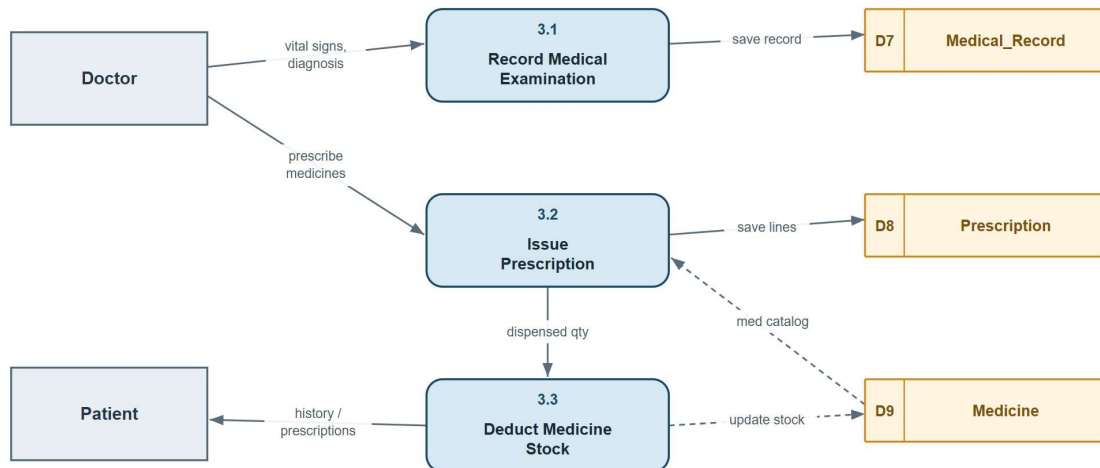


Figure 6: DFD Level 1 – Clinical Examination & Prescription

5.2.4 Billing & Payment

Generates consultation and pharmacy invoices, records details, and processes payments. Data stores: Invoice, Invoice_Detail.

DFD Level 1 — Billing & Payment

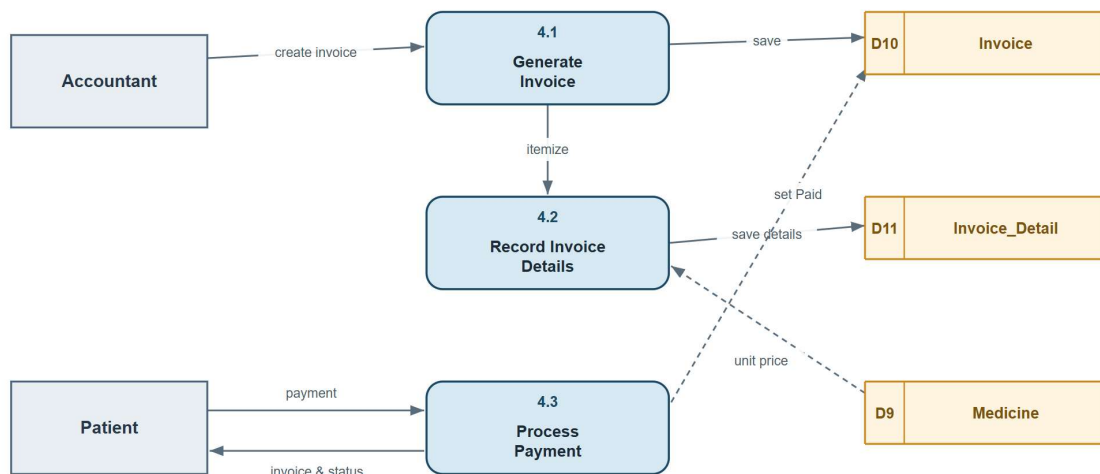


Figure 7: DFD Level 1 – Billing & Payment

5.2.5 CMS & Contact

Manages blog posts and contact inquiries from the website. Data stores: Post, Contact_Inquiry.

DFD Level 1 — CMS & Contact

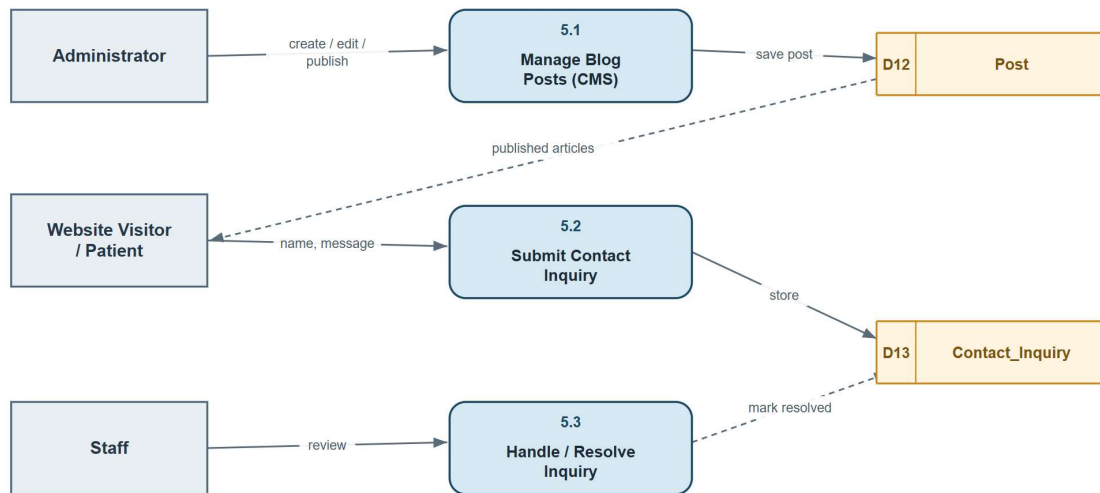


Figure 8: DFD Level 1 – CMS & Contact

5.2.6 Administration & Reporting

Provides user/catalog management and analytics dashboards across all collections. Data stores: all collections.

DFD Level 1 — Administration & Reporting

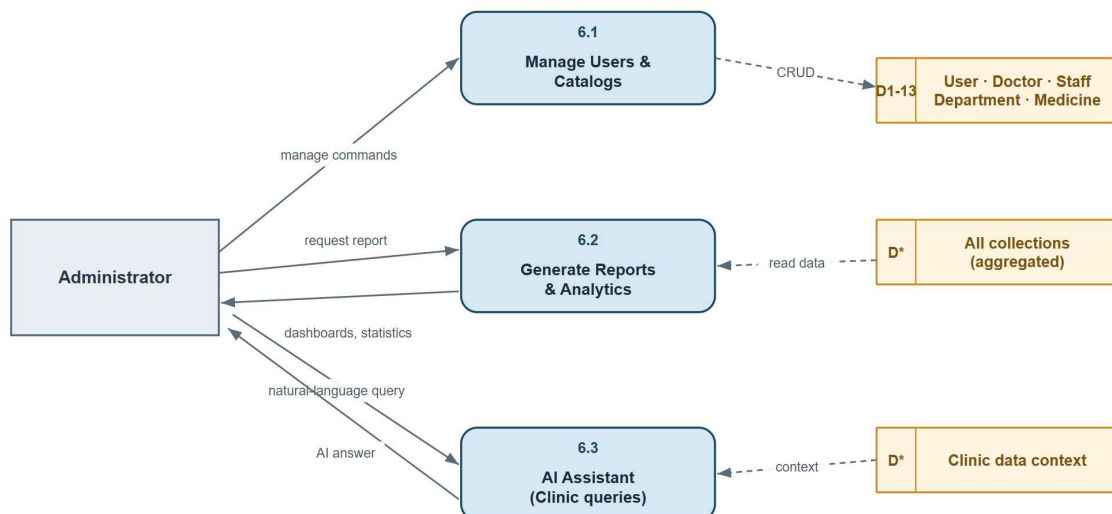


Figure 9: DFD Level 1 – Administration

Chapter 6. INTERFACE OF APPLICATION

This chapter presents the main user interfaces of the Hopsontai Clinic Management System. Screenshots should be inserted in the placeholders below. The interface is organized by area: the public website and the four authenticated role portals.

6.1 Public Home Page

The landing page introduces the clinic with a hero banner, quick-booking widget, featured departments, specialist doctors, service packages, and the latest health articles. Visitors can navigate to Departments, Specialists, Posts, About, and Contact pages.

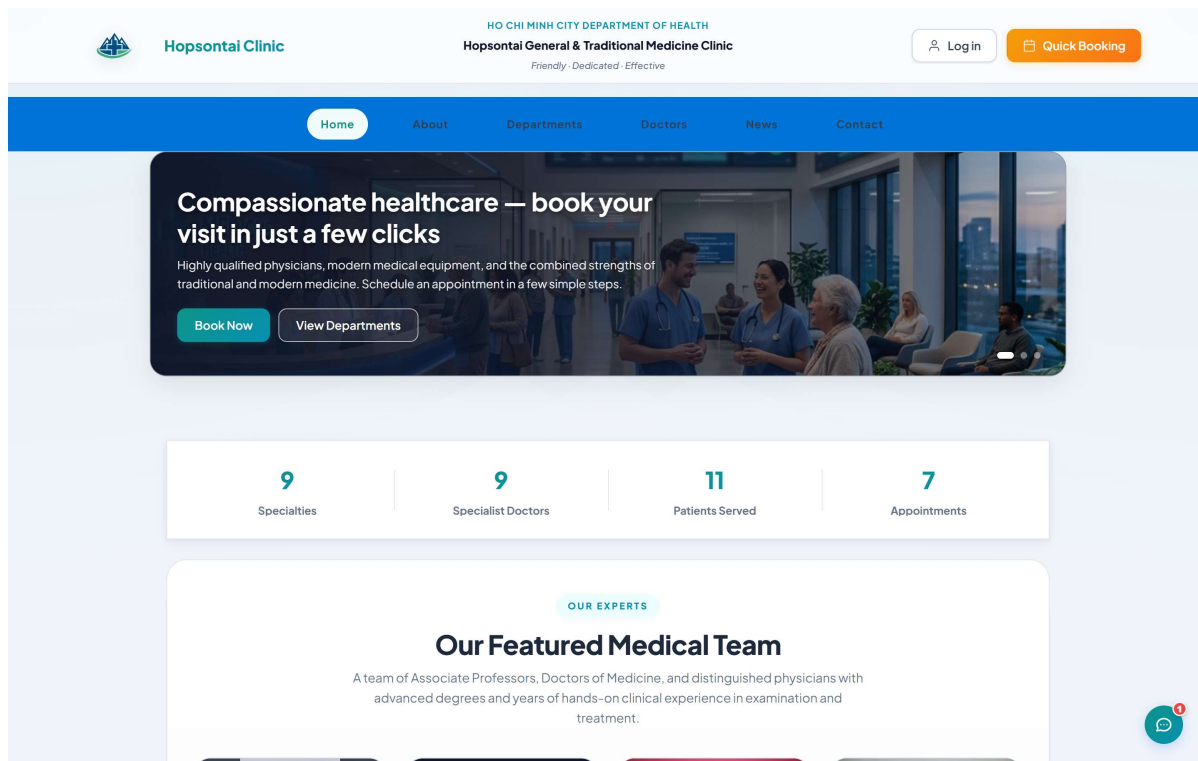


Figure 10: Public home page

6.2 Login Screen

Users log in with their phone number (username) and password. On success, a JWT is stored and the user is redirected to the portal matching their role.

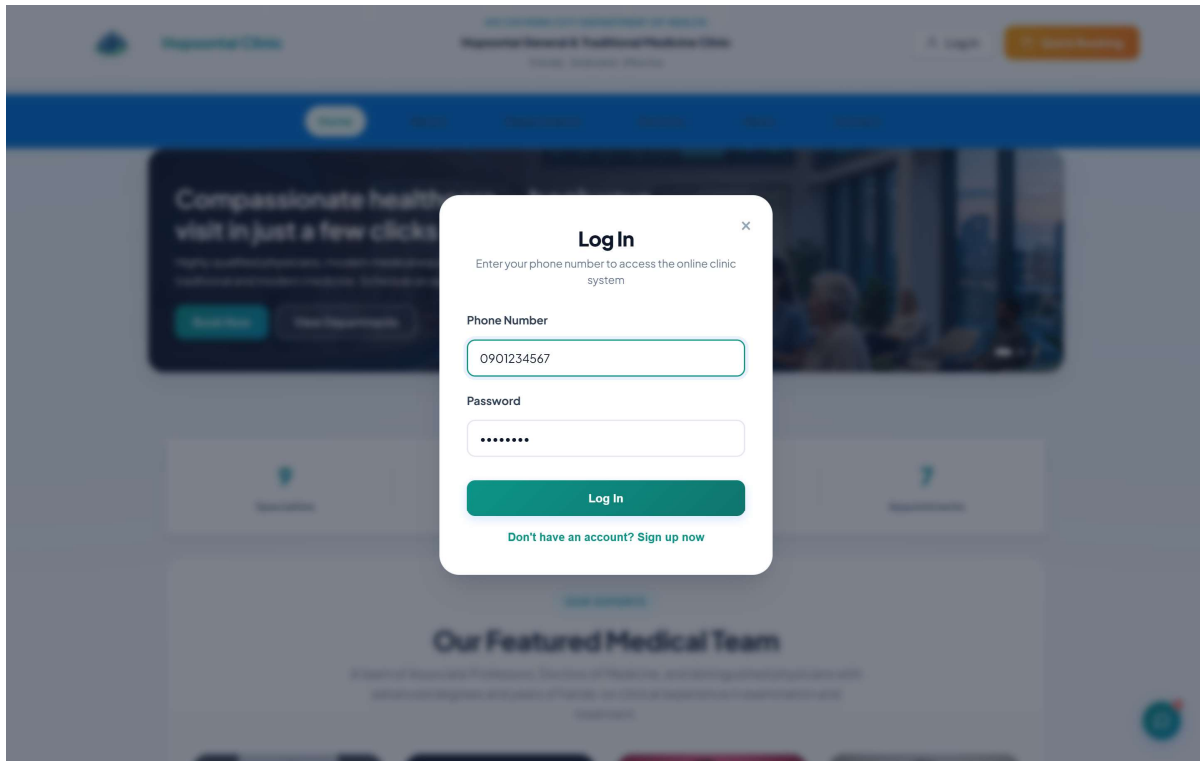


Figure 11: Login screen

6.3 Patient Dashboard & Booking

Patients view upcoming and past appointments, book new appointments through a step-by-step form (department → doctor → date → time slot), and review their medical history, prescriptions, and invoices.

Hopsontai General Clinic
PATIENT

Dashboard Home **LV** Le Van Minh Patient Log out

Failed to load data.

Hello, guest

View your appointments, payments, and medical records at a glance.

QUICK BOOKING

Request a new appointment

Choose a department and a convenient time, then our care team will confirm.

Department

Cardiology

Appointment date

mm/dd/yyyy

Preferred doctor

Select a doctor (optional)

Time slot

08:00

Symptoms / Reason for visit

Recurring headaches and mild fever for the past two days; would like a general check-up.

U

Guest

No phone number

Date of birth ---

Gender ---

ID card ---

Insurance no. ---

Address ---

Book appointment

Appointments

Edit profile

Invoices (0)

Figure 12: Patient dashboard & appointment booking

6.4 Doctor Schedule & Examination

Doctors see their daily schedule and patient queue, open an appointment to record vital signs, diagnosis, and clinical notes, and issue prescriptions from the medicine catalog.

The screenshot shows the interface for a doctor's examination. The top navigation bar includes 'Dashboard', 'Home', and a user profile for 'Assoc. Prof. Dr. Tran Van Hung' with a 'Log out' button. The left sidebar identifies the user as 'Dr. Assoc. Prof. Dr. Tran Van Hung' in 'Cardiology & Intervention' and provides links for 'Patient list', 'Medical records lookup', and 'Work schedule'. The main content area is titled 'Patients to examine today' and includes a sub-header 'View the list of patients confirmed by reception / customer care.' Below this is a table with columns for Patient, Date, Time, Phone, Status, and Actions.

PATIENT	DATE	TIME	PHONE	STATUS	ACTIONS
Le Van Minh DOB: 5/15/1990 Gender: Nam	6/4/2026	10:30	0914444444	Waiting	Start examination
Hoang Minh Tran DOB: 12/10/1995 Gender: Nam	6/1/2026	14:30	0914000002	Examined	Record saved
Le Van Minh DOB: 5/15/1990 Gender: Nam	5/5/2026	09:00	0914444444	Examined	Record saved

Figure 13: Doctor schedule & examination screen

6.5 Staff Customer-Care Dashboard

Customer-care staff confirm or cancel pending appointments and respond to contact inquiries submitted from the public website.

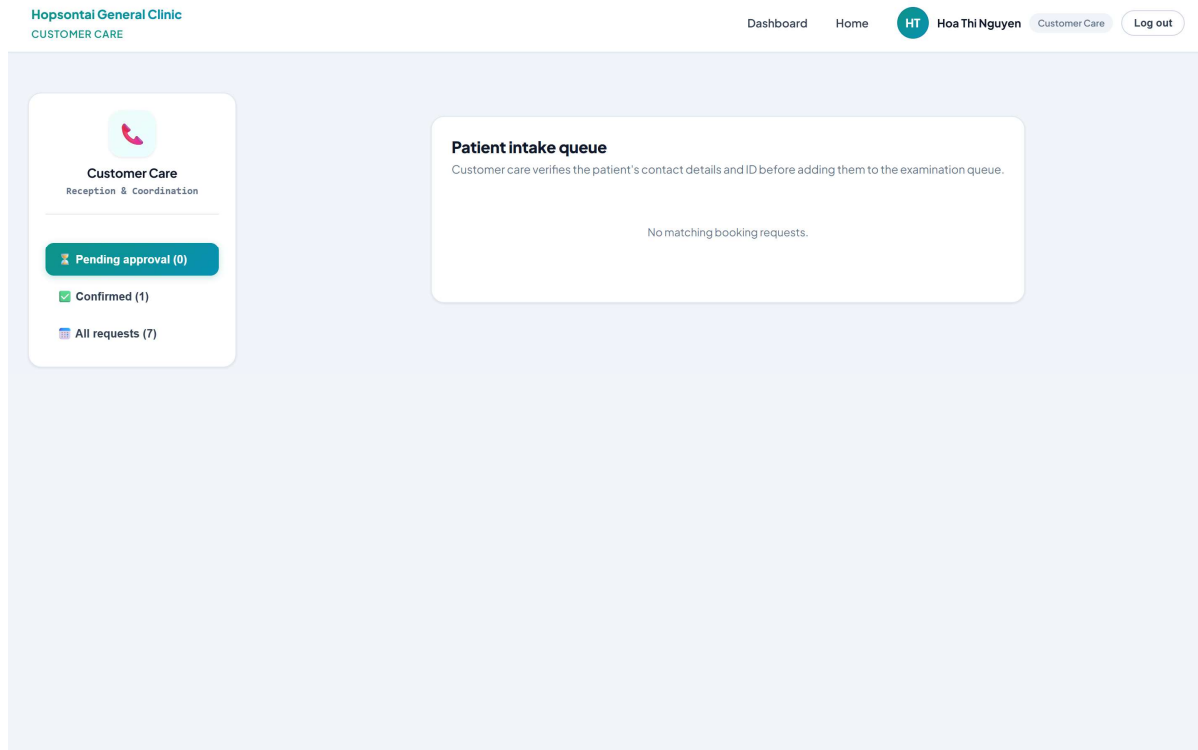


Figure 14: Staff customer-care dashboard

6.6 Accountant Billing Dashboard

Accountants view consultation and pharmacy invoices, process payments, and monitor paid/unpaid status and revenue.

Hopsontai General Clinic
ACCOUNTANT

Dashboard Home **KT** Ke Toan Tran Accountant Log out

Accounting
Cashier & Pharmacy Billing

Hospital fees

Daily revenue report

Search: patient name, phone, or invoice ID...

All invoice types

All statuses

Patient billing management

INVOICE ID	PATIENT	FEE TYPE	TOTAL	ISSUE DATE	STATUS	ACTIONS
E9D117	Mai Thi Vu Phone: 0914000005	Prescription medicine	₫120,000	6/4/2026	Paid	View receipt
E9D115	Mai Thi Vu Phone: 0914000005	Consultation	₫200,000	6/4/2026	Paid	View receipt
E9D186	Tuan Anh Hoang Phone: 0914000004	Prescription medicine	₫24,000	6/3/2026	Paid	View receipt
E9D184	Tuan Anh Hoang Phone: 0914000004	Consultation	₫300,000	6/3/2026	Paid	View receipt
	Huong Thanh					View receipt

Figure 15: Accountant billing dashboard

6.7 Administrator Dashboard

Administrators manage users, departments, doctor schedules, medicines, and blog posts, and view analytics on appointments and revenue. An AI assistant answers clinic-related queries.

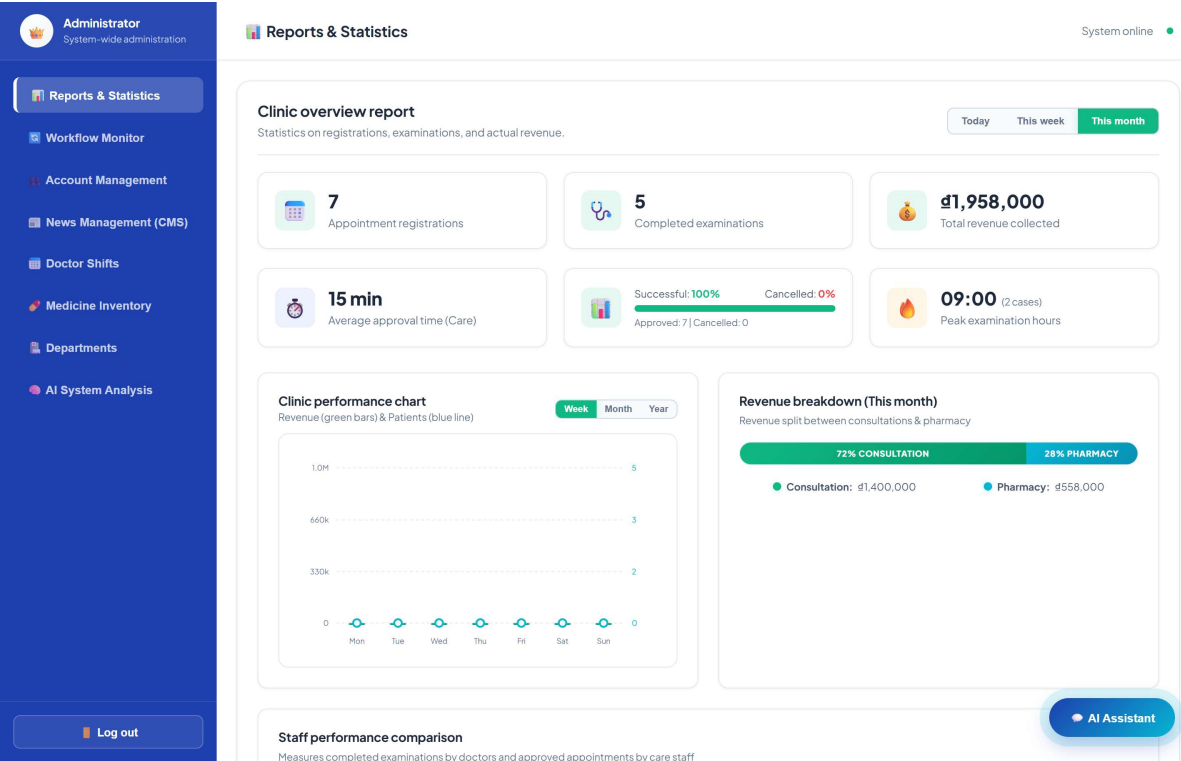


Figure 16: Administrator dashboard

TASK SHEET REVIEW 2

Project Title: Hopsontai Clinic Management System

Date of Preparation of Activity Plan: 13-Jun-2026

Sr. No.	Task	Actual Days	Teammate	Status
1	Database Design – Relational model & collections	3	Nguyen Ngoc Son	Completed
2	Database Design – Data description tables	2	Tran Dang Khoa	Completed
3	DFD Diagrams – Context & Level 1	3	Tran Dang Khoa	Completed
4	Interface of Application – Public & Patient	2	Nguyen Ngoc Son	Completed
5	Interface of Application – Doctor & Staff	2	Nguyen Van Hop	Completed
6	Interface of Application – Accountant & Admin	2	Phan Minh Tai	Completed

Signature of Instructor:

Mrs. Pham Thi Lanh

Signature of Team Leader:

Phan Minh Tai

Chapter 7. USER GUIDE & INSTALLATION

7.1 Installation Guide

Follow these steps to set up the Hopson tai Clinic Management System on your local machine.

a. System Requirements

- Runtime: Node.js v18 or higher, with npm.
- Database: MongoDB (local server or MongoDB Atlas).
- Recommended IDE: Visual Studio Code.
- Tools: Git, MongoDB Compass, and a modern web browser.

b. Backend Setup

1. Open a terminal and navigate to the backend folder: `cd backend`
2. Install dependencies: `npm install`
3. Create a `.env` file with the following variables:
`NODE_ENV=development PORT=4000`
`MONGODB_URI=mongodb://localhost:27017/eproject_clinic`
`JWT_SECRET=your_secret_key FRONTEND_URL=http://localhost:5173`
4. Seed sample data into the database: `node src/config/seed.js --force`
5. Start the backend server: `npm run dev` (API runs at `http://localhost:4000`)

c. Frontend Setup

1. Open a second terminal and navigate to the frontend folder: `cd frontend`
2. Install dependencies: `npm install`
3. Start the development server: `npm run dev`
4. Open the application in a browser at `http://localhost:5173`

d. Test Accounts

After seeding, the following sample accounts are available (username / password):

Role	Username (phone)	Password
Administrator	0901234567	admin123
Doctor	0911111111	doctor123
Staff (Customer Care)	0913333333	staff123
Accountant	0915555555	accountant123
Patient	0914444444	patient123

7.2 User Guide

7.2.1 Patient

The Patient portal lets users manage their healthcare journey online.

- **Login / Register:** Register with your phone number and password, or log in with an existing account.

- **Book an Appointment:** Choose a department, select a specialist doctor, pick an available date and time slot, and describe your symptoms. The appointment is created with status “Pending” until confirmed.
- **My Appointments:** View upcoming and past appointments and their status (Pending, Confirmed, Completed, Canceled).
- **Medical History:** Review previous diagnoses, vital signs, clinical notes, and prescriptions recorded by your doctor.
- **Invoices:** View consultation and pharmacy invoices and their payment status.

7.2.2 Doctor

The Doctor portal supports daily clinical work.

- **Login:** Log in to access your schedule and patient queue.
- **My Schedule:** View work schedules (date, time block, capacity) and the list of confirmed appointments for each session.
- **Examination:** Open a patient’s appointment, record vital signs (height, weight, blood pressure, heart rate, temperature), enter the diagnosis and clinical notes, and save the medical record.
- **Prescription:** Add medicines from the catalog with dosage, frequency, duration, and special instructions. Stock is automatically deducted.
- **Complete Visit:** Mark the appointment as Completed once the examination and prescription are finished.

7.2.3 Staff (Customer Care)

The Staff portal handles front-desk and customer-care tasks.

- **Appointment Confirmation:** Review pending appointments and confirm or cancel them, assigning patients to the appropriate doctor schedule.
- **Contact Inquiries:** Read inquiries submitted from the public Contact page and mark them as resolved once handled.

7.2.4 Accountant

The Accountant portal manages clinic finances.

- **Invoice List:** View all consultation and pharmacy invoices with their status (Unpaid, Paid, Refunded).
- **Create Invoice:** Generate invoices linked to a patient’s appointment.
- **Process Payment:** Record a payment, set the invoice to Paid, and capture the processing date and cashier.

7.2.5 Administrator

The Administrator holds the highest authority and oversees the whole system.

- **Dashboard & Analytics:** Monitor statistics on users, appointments, and revenue, with charts and an AI assistant for clinic queries.
- **User Management:** Create, edit, deactivate, and delete user accounts; provision doctors and staff; impersonate a user for troubleshooting.

- **Department & Schedule Management:** Add and edit departments and create or remove doctor work schedules.
- **Medicine Management:** Maintain the medicine catalog (code, name, ingredient, price, stock).
- **Content Management (CMS):** Create, edit, publish, and delete clinical blog posts, including thumbnail upload.
- **Contact Inquiries:** Review and resolve contact inquiries from the public website.

g. Troubleshooting & Common Scenarios

- **Cannot log in:** Ensure the backend is running on port 4000 and the database has been seeded. Use the exact phone-number username and password.
- **Empty data:** Run the seed script (`node src/config/seed.js --force`) to populate sample departments, doctors, medicines, and posts.
- **Booking unavailable:** A schedule marked “Full” has reached its maximum patient capacity; choose another time slot or date.
- **Access denied (403):** The logged-in role does not have permission for that action – log in with an account that has the required role.

TASK SHEET REVIEW 3

Project Title: Hopsontai Clinic Management System

Date of Preparation of Activity Plan: 20-Jun-2026

Sr. No.	Task	Actual Days	Teammate	Status
1	Installation Guide & Backend setup documentation	2	Nguyen Ngoc Son	Completed
2	User Guide – Patient & Doctor	2	Tran Dang Khoa	Completed
3	User Guide – Staff, Accountant & Admin	2	Nguyen Van Hop	Completed
4	Final review, formatting & report compilation	2	Phan Minh Tai	Completed

Signature of Instructor:

Mrs. Pham Thi Lanh

Signature of Team Leader:

Phan Minh Tai